INCH-POUND

MIL-B-85667/12
28 January 1991

MILITARY SPECIFICATION SHEET

BOLT, SLEEVE, 100° TENSION FLUSH HEAD, STRAIGHT SHANK,
CRES 304 SLEEVE, TITANIUM ALLOY PIN, 95 KSI Fsu, 160 KSI Ftu

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-85667.

1. Application criteria. The sleeve bolt shall be used to provide a precision net fit or controlled interference fit. The sleeve bolts shall be used in accordance with the limitations of MIL-B-85667/1(AS). The sleeve bolts shall not be degreased and cetyl alcohol shall not be removed before installation.

2. Companion and accessory parts. Nuts in accordance with MIL-N-85729 shall be used with the sleeve bolts.

3. Oversize bolts. First and second oversize sleeve bolts may replace MIL-B-85667/12 sleeve bolts for first and second repair operations.

4. Installation. Installation shall be in accordance with MIL-HDBK-271-104.

5. Materials.

5.1 Pin. Pin material shall be titanium 6AL-4V in accordance with AMS 4967 or AMS 4928 (UNS R56400).

5.2 Sleeve. Sleeve material shall be corrosion resistant steel, AISI 304 (UNS S30400), in accordance with MIL-T-8504.

6. Heat treatment.

6.1 Pin. Pin shall be heat treated in accordance with MIL-H-81200 to 95 KSI Fsu minimum and 160 KSI Ftu minimum.

AMSC N/A FSC 5306

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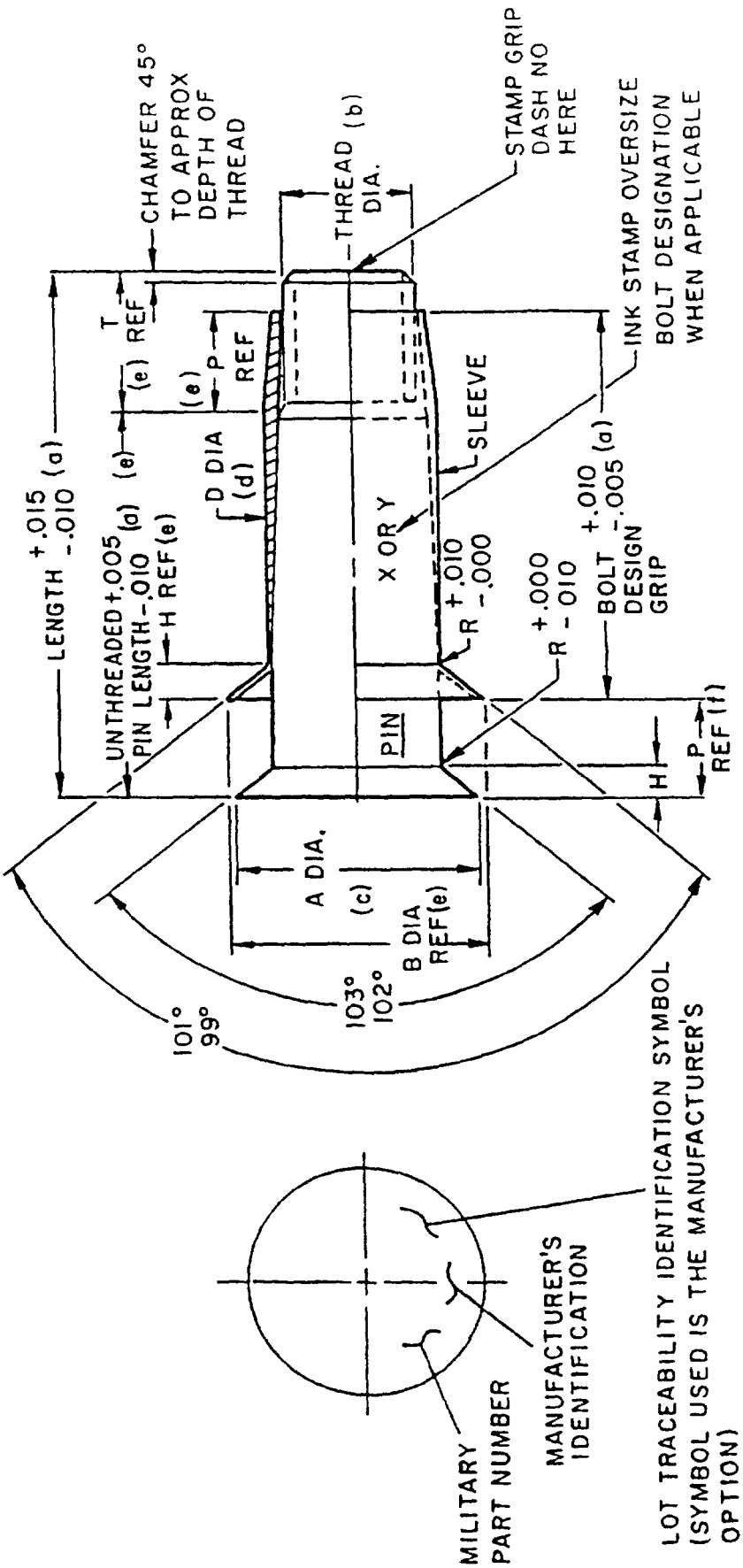


FIGURE 1. Tension flush head bolt.

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NOTES

a. Grip lengths are available in .0625 increments.

b. Threads shall be in accordance with MIL-S-8879. Thread major diameter (T.D.) shall be in accordance with Table I.

c. 1. For sizes .190 thru .375, dimension "A" shall be concentric to dimension "D" within .008 FIM.

2. For sizes .437 thru .625, .012 FIM.

3. For size .750, .020 FIM.

d. 1. For sizes .190 thru .375, dimension "D" shall be concentric to thread pitch diameter within .006 FIM.

2. For sizes .437 thru 0.750, .010 FIM.

e. Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.

f. Protrusion will vary according to taper to provide expansion as given in Table I.

6.2 Sleeve. Sleeve is cold worked.

7. Surface texture. Surface texture shall be R_a-32 in accordance with ANSI/ASME B46.1.

8. Coatings or surface treatment.

8.1 Pin. The pin shall be dry-film lubricated in accordance with MIL-L-46010, Type I, .0003/.0005 thick.

8.2 Sleeve. None.

9. Lubricants. 0.0001-inch overspray allowed on head and threads. After application of dry lubrication, the pin outside diameter and sleeve inside diameter shall be lubricated with cetyl alcohol in accordance with MIL-L-8718.

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TABLE I. Bolt dimensions.

Dash Numbers	Head Per MS 8674 UNJF-3A	Bolt Head Dia	Sleeve Flange Diameter Std	Sleeve Flange Diameter X O/S	Sleeve Flange Diameter Y O/S	Flange Height Ref	Fillet R	Thread Dia	Thread Length	Minimum Sleeve Diameter Std	Minimum Sleeve Diameter X O/S	Minimum Sleeve Diameter Y O/S	Diameter Of Final Sleeve Diameter (Min) Std	Diameter Of Final Sleeve Diameter (Min) X O/S	Diameter Of Final Sleeve Diameter (Min) Y O/S	Installation Hole Size Diameter Std	Installation Hole Size Diameter X O/S	Installation Hole Size Diameter Y O/S
3																		
7 thru 10	1900-32	.418 .369	.457	.463	.469	.093	.225	.030	.1840 .1810	.343								
						.083												
11 thru -16		.420 .370	.459	.465	.471	.093	.432											
-17 thru -22		.430 .380	.476	.483	.490	.091	.432											
4																		
8 thru 10	2500-28	.550 .494	.586	.597	.603	.120	.225	.030	.2440 .2410	.406	.3005 .3000	.3115 .3110	.3175 .3170	.3095 .3085	.3205 .3195	.3265 .3255	.301	.312	.318
						.110												
11 thru 17		.553 .496	.588	.599	.605	.120	.432				.3005 .3000	.3115 .3110	.3175 .3170	.3095 .3085	.3205 .3195	.3265 .3255	.301	.312	.318
-18 thru -28		.566 .515	.607	.614	.623	.108	.432				.3145 .3140	.3215 .3210	.3305 .3300	.3235 .3225	.3305 .3295	.3395 .3385	.315	.327	.331
5																		
-8 thru -11	3125-24	.684 .618	.717	.724	.731	.150	.225	.040	.3060 .3020	.484	.3665 .3660	.3735 .3730	.3805 .3800	.3755 .3745	.3825 .3815	.3895 .3885	.367	.374	.381
						.138												
12 thru -20		.688 .622	.719	.726	.733	.148	.432				.3665 .3660	.3735 .3730	.3805 .3800	.3755 .3745	.3825 .3815	.3895 .3885	.367	.374	.381
-21 thru -31		.702 .636	.740	.748	.755	.136	.432				.3805 .3800	.3885 .3880	.3955 .3950	.3895 .3885	.3975 .3965	.4045 .4035	.381	.389	.396
6																		
-9 thru -11	3750-24	.819 .746	.848	.856	.868	.175	.225	.040	.3680 .3640	.562	.4315 .4310	.4395 .4390	.4515 .4510	.4405 .4395	.4485 .4475	.4605 .4595	.432	.440	.452
						.166												
-12 thru -22		.823 .750	.851	.859	.871	.176	.432				.4315 .4310	.4395 .4390	.4515 .4510	.4405 .4395	.4485 .4475	.4605 .4595	.432	.440	.452
23 thru -34		.838 .765	.872	.880	.888	.164	.432				.4475 .4470	.4555 .4550	.4635 .4630	.4565 .4555	.4645 .4635	.4725 .4715	.448	.456	.464
7																		
-9 thru -12	4375-20	.956 .875	.987	.999	1.007	.205	.225	.050	.4330 .4260	.625	.4985 .4980	.5105 .5100	.5185 .5180	.5075 .5065	.5195 .5185	.5275 .5265	.499	.511	.519
						.194												
-13 thru -26		.960 .878	.991	1.003	1.011	.207	.432				.4985 .4980	.5105 .5100	.5185 .5180	.5075 .5065	.5195 .5185	.5275 .5265	.499	.511	.519
-27 thru -40		.978 .895	1.015	1.026	1.038	.192	.432				.5185 .5180	.5295 .5290	.5415 .5410	.5275 .5265	.5385 .5375	.5505 .5495	.519	.530	.542
8																		
-10 thru -12	5000-20	1.093 1.005	1.120	1.130	1.140	.231	.225	.050	.4930 .4880	.703	.5695 .5690	.5795 .5790	.5895 .5890	.5785 .5775	.5885 .5875	.5985 .5975	.570	.580	.590
						.222												
-13 thru -30		1.097 1.008	1.125	1.135	1.145	.233	.432				.5695 .5690	.5795 .5790	.5895 .5890	.5785 .5775	.5885 .5875	.5985 .5975	.570	.580	.590
-31 thru -46		1.118 1.035	1.152	1.162	1.171	.220	.432				.5895 .5890	.5995 .5990	.6085 .6080	.5985 .5975	.6085 .6075	.6175 .6165	.590	.600	.609
9																		
-14 thru -23	5625-18	1.195 1.100	1.232	1.247	1.260	.249	.432	.070	.5550 .5500	.781	.6395 .6390	.6545 .6540	.6675 .6670	.6485 .6475	.6635 .6625	.6765 .6755	.640	.655	.668
						.240												
-24 thru -48		1.226 1.131	1.274	1.290	1.305	.253	.432				.6705 .6700	.6865 .6860	.7015 .7010	.6795 .6785	.6955 .6945	.7105 .7095	.671	.687	.702
-49 thru -72		1.226 1.131	1.274	1.290	1.305	.238	.900				.6705 .6700	.6865 .6860	.7015 .7010	.6795 .6785	.6955 .6945	.7105 .7095	.671	.687	.702
10																		
-14 thru -25	6250-18	1.298 1.195	1.355	1.374	1.394	.272	.432	.073	.6160 .6120	.843	.7075 .7070	.7265 .7260	.7465 .7460	.7165 .7155	.7355 .7345	.7555 .7545	.708	.727	.747
						.255												
-26 thru -52		1.332 1.231	1.389	1.407	1.422	.270	.432				.7465 .7460	.7645 .7640	.7795 .7790	.7555 .7545	.7735 .7725	.7885 .7875	.747	.765	.780
-53 thru -78		1.332 1.231	1.389	1.407	1.422	.253	.900				.7465 .7460	.7645 .7640	.7795 .7790	.7555 .7545	.7735 .7725	.7885 .7875	.747	.765	.780
12																		
14 thru 29	7500-16	1.492 1.389	1.549	1.564	1.580	.296	.43	.073	.7430 .7370	1.000	.8475 .8470	.8575 .8570	.8735 .8730	.8515 .8505	.8665 .8655	.8825 .8815	.843	.858	.874
						.260												
-30 thru -60		1.531 1.417	1.588	1.608	1.624	.295	.432				.8845 .8840	.9045 .9040	.9205 .9200	.8935 .9125	.9135 .9125	.9295 .9285	.885	.905	.921
61 thru 90		1.531 1.417	1.588	1.608	1.624	.278	.900				.8845 .8840	.9045 .9040	.9205 .9200	.8935 .9125	.9135 .9125	.9295 .9285	.885	.905	.921

NOTES:

(A) D dia shall be verified within the grip bearing portion engaging the bolt shank prior to expansion; when bolt is installed to sleeve flange the dia. shall be within the expanded D dia. Limits shown in Table I.

(B) The sleevebolt is manufactured as a precision assembly and the components should never be dismantled. In the event of inadvertent separation, please discard and replace with a new assembly.

(C) All dimensions in inches, tolerances: decimals ±.010", angles ±2° unless otherwise specified.

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TABLE II. Mechanical properties.

Dia Dash Number	Ultimate Tensile Strength Lbs-Min (b)	Thread Area In^2 (b)	Double Shear Strength Lbs-Min (a)	Pin Test Diameter (+.0005) In. (a)	Pin Test Dia Area $In.^2$ (a)	Max Install Torque In.-Lbs	Fatigue Test Loading Lbs ± 2% (c)	
							Max	Min
-3	3,470	.0217	5,380	.1900	.0284	60	1,600	160
-4	6,210	.0388	9,300	.2500	.0491	115	2,860	285
-5	9,820	.0614	14,600	.3125	.0767	185	4,520	450
-6	15,200	.0950	21,000	.3750	.1104	335	6,990	700
-7	20,600	.1288	28,600	.4375	.1503	540	9,480	950
-8	27,500	.1717	37,300	.5000	.1963	805	12,600	1,260
-9	34,800	.2176	47,200	.5625	.2485	1080	16,000	1,600
-10	43,600	.2724	58,300	.6250	.3068	1560	20,100	2,010
-12	63,200	.3952	83,900	.7500	.4418	2930	29,100	2,910

Notes:

a. Shear test performed on pins ground to pin test diameter. The shear values are based on a minimum shear strength of 95 KSI and the stress area at the test diameter.

b. The ultimate tensile strength values are calculated on the basic pitch diameter of the external thread and a tensile strength of 160 KSI.

c. Maximum fatigue load is based on 46% of the ultimate tensile strength. Minimum fatigue load is 10% of maximum fatigue load.

10. Example of military part number.

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- 34 — Grip length dash number in sixteenths of an inch. (Stamp on threaded end.)
- '-' — designates basic shank diameter: X designates 1st oversize, Y designates 2nd oversize. (Ink stamp on sleeve when applicable.)
- 12 — Diameter dash numbers in sixteenths of an inch.
- M85667/12 — Basic specification sheet number.

11. Administrative and contractual provisions.

a. For design purposes and in case of conflict, this document takes precedence over documents referenced herein.

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TABLE III. Bolt grips and lengths.

Grip Number	Bolt Design Grip	-3 Diameter		-4 Diameter		-5 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
6	.375	.405	.748	.405	.811	.405	.889
7	.438	.468	.811	.468	.874	.468	.952
8	.500	.530	.873	.530	.936	.530	1.014
9	.562	.592	.935	.592	.998	.592	1.076
10	.625	.655	.998	.655	1.061	.655	1.139
11	.688	.718	1.061	.718	1.124	.718	1.202
12	.750	.780	1.123	.780	1.186	.780	1.264
13	.812	.842	1.185	.842	1.248	.842	1.326
14	.875	.905	1.248	.905	1.311	.905	1.389
15	.938	.968	1.311	.968	1.374	.968	1.452
16	1.000	1.030	1.373	1.030	1.436	1.030	1.514
17	1.062	1.092	1.435	1.092	1.498	1.092	1.576
18	1.125	1.155	1.498	1.155	1.561	1.155	1.639
19	1.188	1.218	1.561	1.218	1.624	1.218	1.702
20	1.250	1.280	1.623	1.280	1.686	1.280	1.764
21	1.312	1.234	1.685	1.234	1.748	1.342	1.826
22	1.375	1.405	1.748	1.405	1.811	1.405	1.889
23	1.438	1.468	1.811	1.468	1.874	1.468	1.952
24	1.500	1.530	1.873	1.530	1.936	1.530	2.014
25	1.562			1.592	1.998	1.592	2.076
26	1.625			1.655	2.061	1.655	2.139
27	1.688			1.718	2.124	1.718	2.702
28	1.750			1.780	2.186	1.780	2.264
29	1.812					1.842	2.326
30	1.875					1.905	2.389
31	1.938					1.968	2.452

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TABLE III. Bolt grips and lengths (continued).

Grip Number	Bolt Design Grip	-6 Diameter		-7 Diameter		-8 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
6	.375	.405	.968	.405	1.030	.405	1.108
7	.438	.468	1.030	.468	1.093	.468	1.171
8	.500	.530	1.092	.530	1.155	.530	1.233
9	.562	.592	1.154	.592	1.217	.592	1.295
10	.625	.655	1.217	.655	1.280	.655	1.358
11	.688	.718	1.280	.718	1.343	.718	1.421
12	.750	.780	1.342	.780	1.405	.780	1.483
13	.812	.842	1.404	.842	1.467	.842	1.545
14	.875	.905	1.467	.905	1.530	.905	1.608
15	.938	.968	1.530	.968	1.593	.968	1.671
16	1.000	1.030	1.592	1.030	1.655	1.030	1.733
17	1.062	1.092	1.654	1.092	1.717	1.092	1.795
18	1.125	1.155	1.717	1.155	1.780	1.155	1.858
19	1.188	1.218	1.780	1.218	1.843	1.218	1.921
20	1.250	1.280	1.842	1.280	1.905	1.280	1.983
21	1.312	1.342	1.904	1.342	1.967	1.342	2.045
22	1.375	1.405	1.967	1.405	2.030	1.405	2.108
23	1.438	1.468	2.030	1.468	2.093	1.468	2.171
24	1.500	1.530	2.092	1.530	2.155	1.530	2.233
25	1.562	1.592	2.154	1.592	2.217	1.592	2.295
26	1.625	1.655	2.217	1.655	2.280	1.655	2.358
27	1.688	1.718	2.280	1.718	2.343	1.718	2.421
28	1.750	1.780	2.342	1.780	2.405	1.780	2.483
29	1.812	1.842	2.404	1.842	2.467	1.842	2.545
30	1.875	1.905	2.467	1.905	2.530	1.905	2.608
31	1.938	1.968	2.530	1.968	2.593	1.968	2.671
32	2.000	2.030	2.592	2.030	2.655	2.030	2.733
33	2.062	2.092	2.654	2.092	2.717	2.092	2.795
34	2.125	2.155	2.717	2.155	2.780	2.155	2.858
35	2.188			2.218	2.843	2.218	2.921
36	2.250			2.280	2.905	2.280	2.983
37	2.312			2.342	2.967	2.342	3.045
38	2.375			2.405	3.030	2.405	3.108
39	2.438			2.468	3.093	2.468	3.171
40	2.500			2.530	3.155	2.530	3.233
41	2.562					2.592	3.295
42	2.625					2.655	3.358
43	2.688					2.718	3.421
44	2.750					2.780	3.483
45	2.812					2.842	3.545
46	2.875					2.905	3.608

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TABLE III. Bolt grips and lengths (continued).

Grip Number	-9 Diameter		-10 Diameter		-12 Diameter	
	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
14	.905	1.686	.905	1.748	.905	1.905
15	.968	1.748	.968	1.811	.968	1.968
16	1.030	1.811	1.030	1.873	1.030	2.030
17	1.092	1.874	1.092	1.935	1.092	2.092
18	1.155	1.936	1.155	1.998	1.155	2.155
19	1.218	1.998	1.218	2.061	1.218	2.218
20	1.280	2.061	1.280	2.123	1.280	2.280
21	1.342	2.124	1.342	2.185	1.342	2.342
22	1.405	2.186	1.405	2.248	1.405	2.405
23	1.468	2.248	1.468	2.311	1.468	2.468
24	1.530	2.311	1.530	2.373	1.530	2.530
25	1.592	2.374	1.592	2.435	1.592	2.592
26	1.655	2.436	1.655	2.498	1.655	2.655
27	1.718	2.498	1.718	2.561	1.718	2.718
28	1.780	2.561	1.780	2.623	1.780	2.780
29	1.842	2.624	1.842	2.685	1.842	2.842
30	1.905	2.686	1.905	2.748	1.905	2.905
31	1.968	2.748	1.968	2.811	1.968	2.968
32	2.030	2.811	2.030	2.873	2.030	3.030
33	2.092	2.874	2.092	2.935	2.092	3.092
34	2.155	2.936	2.155	2.998	2.155	3.155
35	2.218	2.998	2.218	3.061	2.218	3.218
36	2.280	3.061	2.280	3.123	2.280	3.280
37	2.342	3.124	2.342	3.185	2.342	3.342
38	2.405	3.186	2.405	3.248	2.405	3.405
39	2.468	3.248	2.468	3.311	2.468	3.468
40	2.530	3.311	2.530	3.373	2.530	3.530
41	2.592	3.374	2.592	3.435	2.592	3.592
42	2.655	3.436	2.655	3.498	2.655	3.655
43	2.718	3.498	2.718	3.561	2.718	3.718
44	2.780	3.561	2.780	3.623	2.780	3.780
45	2.842	3.624	2.842	3.685	2.842	3.842
46	2.905	3.686	2.905	3.748	2.905	3.905
47	2.968	3.748	2.968	3.811	2.968	3.968
48	3.030	3.811	3.030	3.873	3.030	4.030
49	3.092	3.874	3.092	3.935	3.092	4.092
50	3.155	3.896	3.155	3.998	3.155	4.155
51	3.218	3.998	3.218	4.061	3.218	4.218
52	3.280	4.061	3.280	4.123	3.280	4.280

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TABLE III. Bolt grips and lengths (continued).

Grip Number	-9 Diameter		-10 Diameter		-12 Diameter	
	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
53	3.342	4.124	3.342	4.185	3.342	4.342
54	3.405	4.186	3.405	4.248	3.405	4.405
55	3.468	4.248	3.468	4.311	3.468	4.468
56	3.530	4.311	3.630	4.373	3.530	4.350
57	3.592	4.374	3.592	4.435	3.592	4.592
58	3.655	4.436	3.655	4.498	3.655	4.655
59	3.718	4.499	3.718	4.561	3.718	4.718
60	3.780	4.561	3.780	4.623	3.780	4.780
61	3.843	4.624	3.843	4.889	3.843	4.843
62	3.905	4.686	3.905	4.748	3.905	4.905
63	3.968	4.749	3.968	4.811	3.968	4.968
64	4.030	4.811	4.030	4.873	4.030	5.030
65	4.093	4.874	4.093	4.936	4.093	5.093
66	4.155	4.936	4.155	4.998	4.155	5.155
67	4.218	4.999	4.218	5.061	4.218	5.218
68	4.280	5.061	4.280	5.123	4.280	5.280
69	4.343	5.124	4.343	5.186	4.343	5.343
70	4.405	5.186	4.405	5.248	4.405	5.405
71	4.468	5.249	4.468	5.311	4.468	5.468
72	4.530	5.311	4.530	5.373	4.530	5.530
73			4.593	5.434	4.593	5.593
74			4.655	5.498	4.655	5.655
75			4.718	5.561	4.718	5.718
76			4.780	5.623	4.780	5.780
77			4.843	5.686	4.843	5.843
78			4.905	5.748	4.905	5.905
79					4.968	5.968
80					5.030	6.030
81					5.093	6.093
82					5.155	6.155
83					5.218	6.218
84					5.280	6.280
85					5.343	6.343
86					5.405	6.405
87					5.468	6.468
88					5.530	6.530
89					5.593	6.593
90					5.655	6.655

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b. The bolt assemblies identified herein are proprietary products and are covered by the following patents: U.S. 4,087,896 (expiring May 9, 1997), 4,106,036 (expiring July 25, 1997), 4,048,898 (expiring September 20, 1996), Great Britain 1,524,886, Canada 1,023,585; Japan 50-125256; Italy 1,051,593; Israel 515578, and Germany 2,546,641.5. The Government does not have a royalty-free license.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Review.
Air Force - 82

Preparing Activity
Navy - AS
Project No. 5306-1372

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